October 22, 2026

Approximate Supercell Matching

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lattice; reduction; SuperCell; matching

Abstract

A solution is presented for determining the relationship between two unit cells when one is a supercell of the other.

> *Siblings are recognized at a glance, cousins on reflection, and ancestors only by argument. An acquaintance may be mistaken for any of them.*
>
> — Claude (Anthropic, Inc.), 2026

1. Synopsis

A solution is proposed for the problem of best matching of a unit cell to a supercell.

2. Introduction

Santoro *et al.* (1980) enumerated transformation matrices for generating coincidence site lattices (sublattices) from a given unit cell description. They create one exact

representation of a sublattice for each choice of input and transformation. The general solution requires awareness of different choices of unit cells, alternative choices for basis vectors for actual or near degeneracies, and allowance for lattice distortion (for instance, because of replacement on one metal ion with another).

The program NIST*LATTICE (University of California, Los Angeles, 2024; Karen & Mighell, 1990) is similar to CmdLMP3. NIST*LATTICE uses a more obscure matrix generation module and does not have a simple measure of fit, instead relying on error windows on each of the 6 cell parameters.

Andrews & Bernstein (2026*a*) describe a general solution for finding approximate relationships between unit cells that are similar but not necessarily sampled from the identically-same lattice. A user of that software (CmdLMP3) asked if the same could be applied to the determination of unit cells to supercells. Here we extend their method to the case of supercells. (CmdLMP3 is a component of LatticeRepresentationLib; see Section 7.

Figure 1 summarized the operations of Andrews *et al.* (2023) that are used to compare a "reference" unit cell to a "probe" unit cell; the results are returned in the coordinate frame of the reference.

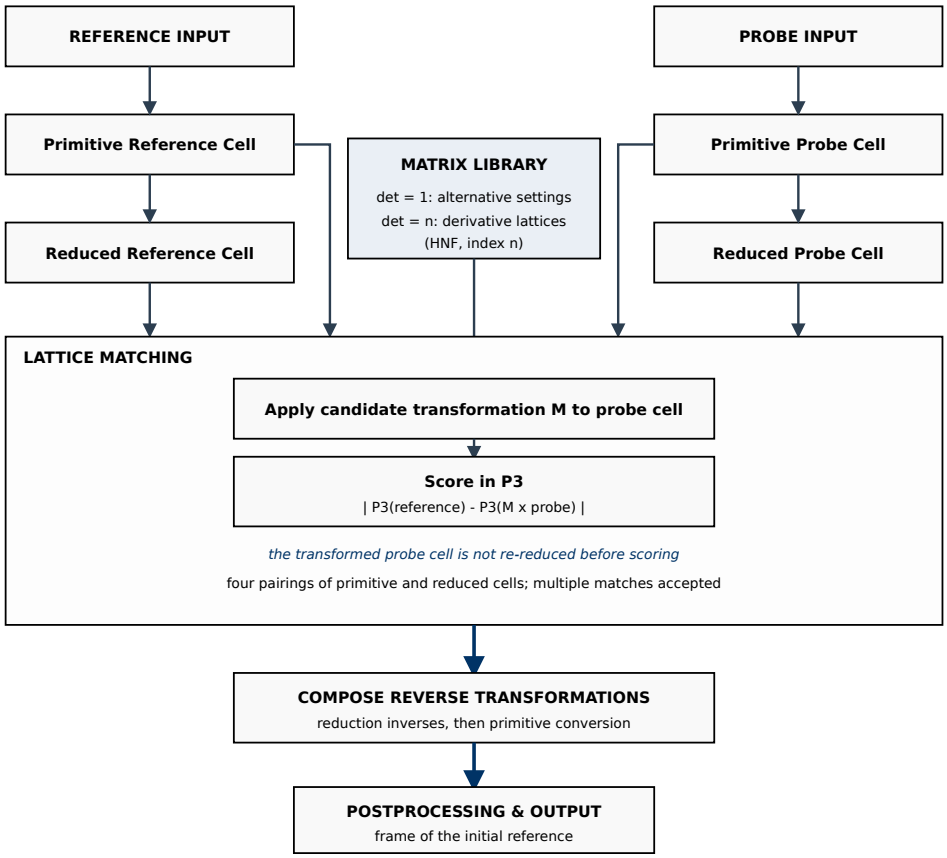


Fig. 1. Flowchart of the approximate lattice matching process of Andrews & Bernstein (2026*a*). Reference and probe cells are converted to primitive form and reduced. Both the primitive and the reduced cell of each are passed to the matching stage, giving four pairings. A library of matrices supplies candidate transformations of the probe: matrices of determinant one enumerate alternative settings of the probe lattice, and matrices of determinant n enumerate the derivative lattices of index n. Each candidate transformation is applied to the probe lattice and scored by the $\mathbf{P}^3$ (Andrews & Bernstein, 2026*b*) distance to the reference. The transformed cell is not reduced again before scoring, so the score is calculated in the coordinate frame of the reference. Accepted matched approximate cells are carried back through the reduction and primitive-conversion inverses to the frame of the initial reference.

3. Algorithm

The algorithm described above and in Figure 1 suffices for matching lattices that are somewhat similar to each other, but the question of matching a cell to a supercell can be tackled with a modest modification: where matching as described in Figure 1 is done using unimodular matrices, by replacing the library of transformation matrices with a library of matrices that have larger determinants, approximate supercell relationships can be determined.

For instance, for a given unit cell, to find supercells that have a base vector tripled a library of transformations with determinant=3.0 would be used.

A library of matrices for a supercell of order n can be constructed by a function that uses 9 nested loops, one for each element of the 3x3 transformation matrices. For transformations of order n, the loops would run from -n to +n, and matrices with determinant = n would be added to the library.

Santoro *et al.* (1980) addressed the question of supercells using a limited set of matrices, the **H**ermite **N**ormal **F**orm (HNF) (Wikipedia contributors, 2026). For a given supercell order (equal to the determinant of the transformation matrix between original and supercell), the HNF gives and one example of each possible lattice. The issues that arise are that the possible exchanges of axes and inexact, measured data are not addressed.

4. Notation

$\mathbf{P}^3$: Polar coordinate space is a compact and smooth alternative to lengths and angles. $\mathbf{P}^3$ is useful for comparing unit cells. (Andrews & Bernstein, 2026*b*)

$\mathbf{P}^3$: $(|\vec{a}|, \alpha),\ (|\vec{b}|, \beta),\ (|\vec{c}|, \gamma)$

$\Rightarrow$

$((|\vec{a}|\cos\alpha,\ |\vec{a}|\sin\alpha),\ \ (|\vec{b}|\cos\beta,\ |\vec{b}|\sin\beta),\ \ (|\vec{c}|\cos\gamma,\ |\vec{c}|\sin\gamma))$

also known as (p_1, p_2, p_3)

5. Implementation details

5.1. Determinants

For each value of determinants (n), the number of possible matrices with maximum elements (-n,n), the number of possible matrices grows rapidly. See Table 1.

Table 1. *Transformation matrix counts by determinant class. For $n = 1$ and $n = 2$ the complete set of integer matrices with determinant = 1 or 2, respectively with elements -1,0,1 or -2,-1,0,1,2, respectively is used in CmdLMP3. For index $n = 5$ and index $n = 6$ the exhaustive search is not attempted, as it would screen 2.4×10^9 and 1.1×10^{10} candidates to yield sets of 1.4×10^7 and 7.1×10^7 matrices. In their place the 3480 matrices of determinant one are composed by multiplication of the Hermite normal forms of index 5 and 6. The resulting sets span all 31 and all 91 derivative lattices, but sample the settings of each lattice less densely than the exhaustive sets at lower index.*

n	Derivative lattices†	Candidates screened‡	Complete set§	Used in CmdLMP3
1	1	19 683	3 480	3 480
2	7	1 953 125	147 696	147 696
3	13	40 353 607	1 060 920	1 060 920
4	35	387 420 489	6 721 896	6 721 896
5	31	2 357 947 691	13 942 248	215 760
6	91	10 604 499 373	70 632 144	633 360
Total	177		92 508 384	8 783 160

† Number of distinct sublattices of index n in a three-dimensional lattice. Each has exactly one Hermite normal form..

‡ Size of the exhaustive search over integer matrices with coefficients in $[-n, n]$, the smallest symmetric box that contains every Hermite normal form of index n. Screening costs are not additive across independent searches, so no total is given.

§ Number of those candidates having determinant $+n$. Matrices of negative determinant are excluded because they invert chirality.

¶ Number of matrices actually retained by CmdLMP3.

5.2. Reference vs. probe decision

For a given input pair of unit cells, the decision to search for supercells of the reference cell requires that the reference have the smaller primitive unit volume of the pair. When 2 input cells are compared and the ratio of he volumes is not near

1.0, then the larger of the 2 is designated as the reference cell and the other as the probe. In the case where the reference cell has the smaller primitive volume, then the reference and probe cells are interchanged.

The output of the results of matching shows the transformation from the input probe cell to the frame of the input reference cells. That might be the reverse of the choice that the user desires, so the inverse 3x3 matrix is also output.

6. Examples

A small selection of example executions of CmdLMP3 is shown. The example range from simple comparisons to a variety of different types of supercells. For the cases of supercells, the program uses the $\mathbf{P}^3$ fit value to attempt to describe the fit. The categories EXCELLENT, GOOD, POOR, and DISTORTED are heuristic labels intended only as a qualitative guide for interpreting the $\mathbf{P}^3$ fit. Thresholds of 0.5%, 2.0%, and 8.0% were chosen empirically based on experience with a large number of test cases and do not represent statistically derived or physically significant boundaries. The $\mathbf{P}^3$ fit is computed from the difference between the transformed probe vector and the reference vector divided by the length of the reference vector. The $\mathbf{S}^6$ angle is also output as a secondary measure of fit; it is the angle between the transformed probe vector and the reference vector.

6.1. Example 1

A case where supercell search is not needed. This is the simplest example of using CmdLMP3. Because the reduced cell volumes are compared, CmdLMP3 does not perform a supercell alignment. 2 possible alignments of the lattices are found.

```
=== TEST: (Charles Simmons, personal communcaion) ===
Description: Nearly identical cells with known transformation
Input cells:
Cell 1 (P):    10.250    10.740    21.080    87.720    75.970    61.530
```

```
Cell 2 (P):    10.250    10.740    21.080    78.960    75.970    61.490
; primitive cell volume ratio, mobile/reference
;  ratio = 1.000

; === LATTICE MATCHING RESULTS ===

; --- Match 1 ---
; EQUIVALENT  P3 distance: 0.010 (0.0% of P3 norm)  S6 angle: 0.01 deg
; Matrix:  [-1 0 0   -1 1 0   0 0 -1] (Determinant: 1.000)
; Inverse: [-1 0 0   -1 1 0   0 0 -1]
; P    10.250    10.740    21.080    87.720    75.970    61.530  [REFERENCE]
; P    10.250    10.739    21.080    87.714    75.970    61.502  [TRANSFORMED MOBILE (CONVENTIONAL)]

; --- Match 2 ---
; EQUIVALENT  P3 distance: 0.023 (0.1% of P3 norm)  S6 angle: 0.03 deg
; Matrix:  [-1 0 0   0 -1 0   -1 0 1] (Determinant: 1.000)
; Inverse: [-1 0 0   0 -1 0   -1 0 1]
; P    10.250    10.740    21.080    87.720    75.970    61.530  [REFERENCE]
; P    10.250    10.740    21.087    87.674    75.893    61.490  [TRANSFORMED MOBILE (CONVENTIONAL)]
```

6.2. Example 2

This example is a synthetic example of a cubic unit cell where a face diagonal has been doubled. As expected in such a high symmetry case, multiple matches are found.

```
 === TEST: FaceDiagonalOrder2 ===
 Description: [SUPERCELL] Cubic doubled along face diagonal -- order-2, dist=0
 Input cells:
 Cell 1 (P):    10.000    10.000    10.000    90.000    90.000    90.000
 Cell 2 (P):    14.142    14.142    10.000    90.000    90.000    90.000
 ; primitive cell volume ratio, mobile/reference
 ;  ratio = 2.000   swapped reference and probe cells

 ; === LATTICE MATCHING RESULTS ===
 ; === SUPERCELL ===  prim. cell volume ratio = 2.000

 ; --- Match 1 ---
 ; order-2 supercell  EXCELLENT  P3 distance: 0.000 (0.0% of P3 norm)  S6 angle: 0.00 deg
 ; Matrix:  [1 -1 0   1 1 0   0 0 1] (Determinant: 2.000  [SUPERCELL])
 ; Inverse: [1/2 1/2 0   -1/2 1/2 0   0 0 1]
 ; P    14.142    14.142    10.000    90.000    90.000    90.000  [REFERENCE]
 ; P    14.142    14.142    10.000    90.000    90.000    90.000  [TRANSFORMED MOBILE (CONVENTIONAL)]

 ; --- Match 2 ---
 ; order-2 supercell  EXCELLENT  P3 distance: 0.000 (0.0% of P3 norm)  S6 angle: 0.00 deg
 ; Matrix:  [1 1 0   -1 1 0   0 0 1] (Determinant: 2.000  [SUPERCELL])
 ; Inverse: [1/2 -1/2 0   1/2 1/2 0   0 0 1]
 ; P    14.142    14.142    10.000    90.000    90.000    90.000  [REFERENCE]
 ; P    14.142    14.142    10.000    90.000    90.000    90.000  [TRANSFORMED MOBILE (CONVENTIONAL)]
```

6.3. Example 3

A case of quadrupling the reference cell in more than one direction with significant distortion (fit labeled "POOR"). $[Zn(XeF_2)_6][AsF_6]_2$ (Clough *et al.*, 2026)

```
=== TEST: (Simon Parsons, personal communication) ===
Description: [SUPERCELL] R primitive vs P triclinic -- order-4, dist=0.820
Input cells:
Cell 1 (P):     9.477     9.477     9.477    64.240    64.240    64.240
Cell 2 (P):     9.482    13.705    20.048    89.510    88.720    86.960
; primitive cell volume ratio, mobile/reference
;  ratio = 3.953  swapped reference and probe cells

; === LATTICE MATCHING RESULTS ===
; === SUPERCELL ===  prim. cell volume ratio = 4.000

; --- Match 1 ---
; order-4 supercell  POOR  P3 distance: 0.820 (3.1% of P3 norm)  S6 angle: 1.16 deg
; Matrix:  [0 1 0   -1 1 -1   -2 0 2] (Determinant: 4.000  [SUPERCELL])
; Inverse: [1/2 -1/2 -1/4   1 0 0   1/2 -1/2 1/4]
; NOTE: P3 distance (3.146% of P3 norm) exceeds threshold 2.000% -- match is distorted
; P     9.482    13.705    20.048    89.510    88.720    86.960  [REFERENCE]
; P     9.477    13.834    20.155    90.000    90.000    84.859  [TRANSFORMED MOBILE]
```

6.4. Example 4

(Parsons, 2026) supplies low and high pressure measurments of $CuSO_4 \cdot 5H_2O$. The experimental orientation matrices were used to compute a trial supercell relationship (which agreed with CmdLMP3). Again, the suggested fit quality is "POOR" due to significant distortion.

```
[   0.996436 -0.992217  0.004600 ]
[  -0.953339 -0.987698 -0.005615 ]
[  -0.006338  0.029284 -1.004780 ]
```

```
; === P3 LATTICE MATCHING ===
; Enter control variables and input vectors (type 'end' to finish):
P 5.8699    5.9614   10.0883   76.9978   82.8643   72.3787 (1 atm)
P 6.9425    9.2826   10.1032   77.6881   86.0909   88.5508 (5-6 GPa)
end
; === PROCESSING INPUT LIST ===
; primitive cell volume ratio, mobile/reference
;  ratio = 1.939   swapped
; Note: input order swapped -- the larger-volume cell is used as reference.
;       The matrix shown transforms the MOBILE (smaller) cell to match the REFERENCE (larger).
```

```
; === LATTICE MATCHING RESULTS ===
; === SUPERCELL ===  prim. cell volume ratio = 2.000

; --- Match 1 ---
; order-2 supercell  POOR  P3 distance: 0.340 (2.2% of P3 norm)  S6 angle: 0.99 deg
; Matrix:  [1 -1 0   -1 -1 0   0 0 -1] (Determinant: 2.000  [SUPERCELL])
; Inverse: [1/2 -1/2 0   -1/2 -1/2 0   0 0 -1]
; NOTE: P3 distance (2.213% of P3 norm) exceeds threshold 2.000% -- match is distorted
; P     6.942     9.283    10.103    77.688    86.091    88.551  [REFERENCE]
; P     6.986     9.549    10.088    77.478    84.974    89.070  [TRANSFORMED MOBILE]
```

7. Availability of code

The code for Lattice Matching (CmdLMP3) and related software tools is available in github.com, in `https://github.com/duck10/LatticeRepLib.git`. The program CmdLMP3 uses the required files.

8. Supporting information

CmdLP3 has two functions in addition to the processing of users' inputs. Their results are included as supporting information. Two text files are deposited. `CmdLMP_HELP_2026_07_18.txt` contains descriptions of possible input parameters that can be used in addition to experimental data input. For instance, "DETAILS" will trigger output of information about intermediate calculations. `CmdLMP3_TEST_2026_07_18.txt` contains about 50 example cases that the command "TEST" causes to be executed.

9. Artificial intelligence notice

Figure 1 was created with the assistance of the AI program Claude distributed by Anthropic, Inc.

10. Acknowledgements

Careful copy-editing and corrections by Frances C. Bernstein are gratefully acknowledged. C. Simmons and S. Parsons kindly supplied test examples.

Funding information Funding for this research was provided in part by: US Department of Energy Offices of Biological and Environmental Research and of Basic Energy Sciences (contract No. DE-AC02-98CH10886; contract No. E-SC0012704); U.S. National Institutes of Health (grant No. P41RR012408; grant No. P41GM103473; grant No. P41GM111244; grant No. R01GM117126, grant No. R24GM154040 (from U.S.NIHGMS, Lawrence Berkeley Laboratory, for DIALS) grant No. 1R21GM129570); Dectris, Ltd.

Portions of the C++ code used in the implementation of CmdLMP3 were generated with the assistance of Claude (Anthropic), a large language model. All AI-generated code was reviewed, tested, and verified by the authors for correctness and adherence to the algorithmic specifications described in this paper; the authors take full responsibility for its accuracy and behavior.

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